

# Assessing the Feasibility of Autonomous Drone Delivery Systems - A Strategic Flexibility Approach

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## 1 Abstract

This report evaluates the economic viability of an autonomous drone delivery system for on-demand food transport in London. The first part of the analysis focuses on three distinct drone hub configurations, each varying in the number and size of drones, assessed through a Discounted Cash Flow (DCF) model. Among the alternatives, a hub with a high number of smaller drones emerged as the most cost-effective option based on financial metrics in a deterministic scenario.

However, this approach and other existing studies do not account for real-world uncertainties. This report addresses this by simulating scenarios designed to compare the economic performance of different designs under varying conditions, including fluctuations in demand, battery costs, and delivery disruptions due to weather. Furthermore, conservative, pragmatic, and risky strategic flexibility approaches using decision rules were introduced to further mitigate risks and enhance system performance. The results indicate that prioritising a hub with a large fleet of small drones reduces initial investment while maximising financial returns, making it the most economically viable configuration.

## 2 Introduction

The rapid expansion of e-commerce and on-demand food delivery has intensified urban logistics challenges in metropolitan areas like London. Traditional ground-based delivery methods contribute significantly to traffic congestion and environmental degradation.

For instance, in 2024, London was ranked as Europe's most congested city, with drivers spending an average of 101 hours in traffic, a 2% increase from the previous year [1], resulting in an estimated £3.85 billion economic cost, or £942 per driver. This congestion is further exacerbated by freight and delivery vehicles, which account for 17%

of London's road traffic [2]. With personal deliveries projected to double by 2030, the reliance on petrol and diesel vans is expected to rise significantly, potentially increasing last-mile delivery emissions by 32% if current practices remain unchanged [3][4].

Autonomous drone delivery systems offer a promising alternative, potentially alleviating congestion and reducing environmental impact. However, their implementation faces uncertainties, including regulatory constraints, demand variability, battery limitations, and weather-related challenges. Incorporating strategic flexibility into the design of these systems enables adaptive responses to such uncertainties, enhancing economic viability and resilience.

This study examines various drone hub configurations for food delivery in London, assessing their financial feasibility under uncertain conditions. Through Discounted Cash Flow (DCF) analysis and scenario simulations, we evaluate how factors like fleet size, battery efficiency, and weather disruptions influence system performance. By integrating flexibility strategies into the design process, we aim to identify the most cost-effective and scalable approach to autonomous food delivery, ensuring adaptability in a rapidly evolving market.

This report aims therefore to answer the following questions:

1. Is a large direct upfront cost expenditure in developing a high-complexity system mitigated by the likelihood of future economic upside?
2. How might we systematically account for the increased uncertainty and risk associated with the deployment of a costly drone network? Can we capture the value of flexibility in our decision-making?

|                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Strengths</b></p> <ul style="list-style-type: none"> <li>· Fast and efficient delivery with reduced congestion and emissions.</li> <li>· Highly scalable for various regions and industries.</li> </ul> | <p><b>Weaknesses</b></p> <ul style="list-style-type: none"> <li>· High upfront costs for infrastructure and technology.</li> <li>· System performance depends on battery efficiency and airspace regulations.</li> </ul>                                            |
| <p><b>Opportunities</b></p> <ul style="list-style-type: none"> <li>· Potential partnerships with major logistics companies to expand reach and enhance operations.</li> </ul>                                 | <p><b>Threats</b></p> <ul style="list-style-type: none"> <li>· Regulatory and safety restrictions may limit deployment.</li> <li>· Uncertain market demand could impact profitability.</li> <li>· Weather conditions may affect availability of service.</li> </ul> |

**Figure 1:** *SWOT analysis of autonomous drone delivery systems*

## 3 Background

The use of autonomous drones for delivery has gained significant traction in recent years, with companies like Amazon Prime Air, Wing (Alphabet), and UPS Flight Forward pioneering pilot programs to assess feasibility [5]. These projects focus on reducing delivery times, minimizing congestion, and lowering emissions, but their large-scale implementation remains limited due to regulatory, technological, and economic challenges.

In terms of project evaluation and system design, prior studies have employed deterministic models to assess drone delivery economics. For example, DCF analysis and Multi-Criteria Decision Analysis (MCDA) have been used to evaluate fixed fleet sizes, operational costs, and revenue potential under stable conditions [6][7]. However, these models often overlook key uncertainties, such as fluctuations in demand, battery degradation, and airspace restrictions, which significantly impact long-term profitability.

Emerging technologies like battery advancements, AI-driven route optimisation, and urban air mobility regulations introduce uncertainties that significantly impact costs and operations but are often overlooked in existing studies. Additionally, gaps in risk assessments, such as weather disruptions, cybersecurity threats, and regulatory changes limit accurate economic forecasting.

By integrating flexibility analysis and stochastic modelling to address these uncertainties, this study aims to improve traditional economic evaluations and proposes more resilient and adaptable drone delivery strategies.

## 4 Methodology

The methodology of this report can be divided into four stages:

1. System Definition
2. Base Case Analysis
3. Uncertainty Analysis
4. Flexibility Analysis

## 4.1 System Definition

The drone delivery network operates using the following simplified architecture, adapted from the Wing drone delivery system [8]:

1. When an order is placed through the system, a drone is dispatched from the hub (where drones recharge and are maintained), or rerouted if they are already flying, to the pick-up site (restaurant, supermarket, etc...)
2. Once arrived, the drone hovers above the site a rope descends from the drone, allowing the pick-up site to fix the payload onto it
3. The secured payload is hoisted up into the air, into the drone, before flying to its drop-off site, where the package is hoisted down for final hand-off



**Figure 2:** *A Wing drone delivering food*

The drone delivery network consists of three alternative fleet configurations, each designed to optimise cost, efficiency and operational performance. Each system includes drone hardware and deployment, excluding software development, regulatory compliance and personnel costs.

The three proposed configurations differ in drone size, fleet quantity, payload capacity and range, affecting cost structures and economic feasibility. Each fleet is housed in a designated hub, incurring land and operational

costs. The system is designed to perform approximately 400 daily deliveries, with each drone completing a set number of trips per day.



**Figure 3:** *A Wing drone hub*

The evaluation considers key performance metrics, including drone acquisition costs, maintenance, battery efficiency, electricity consumption, insurance, and infrastructure expenses. A financial model assesses which configuration yields the highest return on investment while balancing uncertainties in demand, costs, and operational risks.

### 4.1.1 Design Variables

Table 1 and Table 2 summarise the key attributes and cost factors for each fleet configuration.

In order to construct the economic scenarios, a number of estimations and assumptions were made based on market research and averaging results obtained from online drone cost calculation tools [9] [10] [11]. These include:

- **Drone and battery characteristics:** Estimated using price data from leading drone manufacturers (DJI, Parrot), assuming a 5% discount due to bulk purchasing
- **Maintenance costs:** Estimated using the online calculators
- **Installation costs:** As above
- **Land costs:** Estimated using the average price per square meter of a retail space in London (£67), and the footprint of the drone fleet

- **Labour costs:** Estimated a function of the number of drones per fleet, assuming an employee average salary of £30 000, and can operate 3 drones
- **Insurance costs:** Estimated as 10% of the direct upfront costs

Demand and revenue were estimated by synthesising data from market leaders in food delivery in the UK, including Deliveroo [12] and Uber Eats [13].

**Table 1:** *Fleet configuration and Performance metrics*

| Feature                | Small drones fleet | Medium drones fleet | Large drones fleet |
|------------------------|--------------------|---------------------|--------------------|
| Number of drones       | 22                 | 18                  | 15                 |
| Drone Price (£)        | 8,500              | 10,000              | 12,000             |
| Weight (kg)            | 3                  | 6                   | 15                 |
| Payload Capacity (kg)  | 2                  | 5                   | 8                  |
| Range (km)             | 10                 | 15                  | 25                 |
| Battery Type           | Li-Po 4S 5000mAh   | Li-ion 6S 8000mAh   | Li-Po 4S 5000mAh   |
| Battery Life (Ah)      | 5                  | 8                   | 12                 |
| Charging Time (h)      | 1                  | 1.6                 | 2.4                |
| Total Daily Deliveries | 10950              | 14600               | 18250              |

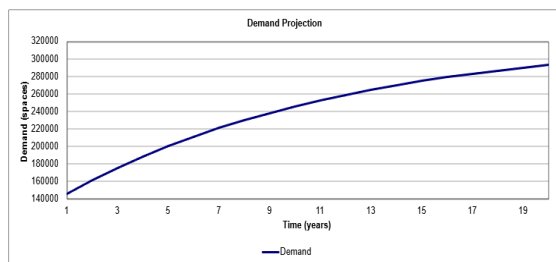
**Table 2:** *Infrastructure and Operating Costs*

| Cost Factor                            | Small drones fleet | Medium drones fleet | Large drones fleet |
|----------------------------------------|--------------------|---------------------|--------------------|
| Battery Charge Cost per year per drone | 37.54              | 52.79               | 15.64              |
| Total Battery Cost                     | 119,98             | 758                 | 1,508              |
| Maintenance per Drone                  | 419,98             | 1,058               | 1,808              |
| Insurance per Drone                    | 400                | 1,000               | 2,300              |
| Labour Cost per Drone                  | 10,000             | 10,000              | 10,000             |
| Hub Land Cost per Year                 | 281,050            | 357,408             | 446,760            |
| Installation Costs                     | 1,000              | 1,000               | 1,000              |

## 5 Deterministic Analysis

The first step of the analysis was to construct a deterministic economic scenario to serve as a base case for further analysis. Certain variables were unchanged across the small, medium and large scenarios to ensure a meaningful comparison. Specifically, demand was assumed to be 400 deliveries per day, given the smaller scale of this “startup”. To meet this demand, a fleet of 22 small drones, 18 medium drones, and 15 large drones was selected.”.

Then, we assumed an average growth rate of 6% per year for the first 10 years. This estimate is based on an initial growth rate of approximately 3% in the early years, reflecting the gradual adoption of new technology. As the technology becomes more established, the growth rate is expected to accelerate to around 8% in later years, resulting in an overall average of 6% over the decade. After this period, the growth rate is assumed to stabilise at 5% for the following 10 years (see Figure 4 below). The results of the subsequent Discounted Cash Flow (DCF) model for each fleet over a defined operational lifespan are summarised below.



**Figure 4:** Demand projection of deliveries

Lastly, we assumed a constant MARR of 12%, given that the core technology of autonomous drone networks has been proven and successfully launched to market.

Each alternative was evaluated using three key financial metrics: Net Present Value (NPV),

Internal Rate of Return (IRR), and Payback Period, providing a comparative assessment of cost-effectiveness and profitability under deterministic conditions.

It is important to acknowledge that while the following analysis supports the financial viability of the different alternatives, the models presented are simplified representations. Their alignment with real-world outcomes depends significantly on the accuracy of the underlying assumptions.

### 5.1.1 Small Drones - Andria

We first considered the large hub configuration of small drones, with DCF informed by the assumptions detailed above. Alternative A requires the most dispersed deployment of drones, as each drone can only carry one delivery at a time due to its small payload capacity and limited range. This means that Alternative A includes the most number of drones of 22.

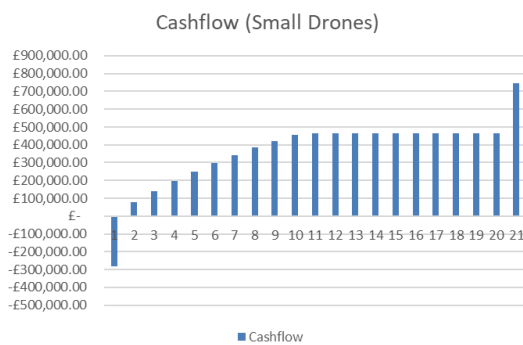
**Table 3:** Alt A Cost Parameters

| Cost                                                         | Value              |
|--------------------------------------------------------------|--------------------|
| Individual Drone Manufacture & Installation Cost (per drone) | £9,500.00          |
| <b>Capacity cost (per drone at 115%)</b>                     | <b>£10,925.00</b>  |
| <b>Annual Infrastructure Costs</b>                           | <b>£281,050.00</b> |
| Yearly Battery Charge (per drone)                            | £37.50             |
| Maintenance (per drone)                                      | £419.98            |
| Insurance / Communication (per drone)                        | £400.00            |
| Total Battery Cost (per drone)                               | £119.98            |
| Labour (per drone)                                           | £10,000            |
| <b>Operating Costs (per drone)</b>                           | <b>£10,977.50</b>  |

As a result of their small, compact size, each drone for Alternative A incurs the lowest

manufacturing and operating costs. Although, manufacturing and operating expenses would likely be optimised over time, for simplicity, these costs are assumed to remain constant for this analysis. With an initial fleet investment, alongside infrastructure and operational costs, the total upfront expenditure per year for this configuration is estimated at £806,905 with year 0 however excluding operational costs.

Using the determined values and predicted demand projection, a deterministic DCF analysis was conducted. The NPV was determined to be £1,476,739.07, with an IRR of 39.21%. This aligns closely with the current market competitors [15]. With the chosen MARR of 12%, Alternative A presents a satisfactory return on investment.



**Figure 5:** Cumulative Cash Flow for Alt A

### 5.1.2 Medium Drones - Arancha

Alternative B offers a balanced approach, featuring fewer units than the small drone fleet, while maintaining higher payload capacity and extended range. This configuration is characterised by moderate acquisition costs, with a total fleet investment of £11,000 per-drone costs. Additionally, infrastructure and operational expenses amount to £12,868.79, contributing to the overall economic assessment of this alternative.

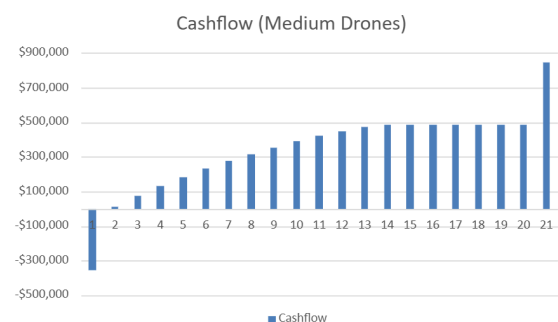
**Table 4:** Alt B Cost Parameters

| Cost                           | Value      |
|--------------------------------|------------|
| Individual Drone Manufacture & | £11,000.00 |

|                                          |                    |
|------------------------------------------|--------------------|
| Installation Cost                        |                    |
| <b>Capacity cost (per drone at 115%)</b> | <b>£12,650.00</b>  |
| <b>Annual Infrastructure Costs</b>       | <b>£357,408.00</b> |
| Yearly Battery Charge (per drone)        | £52.79             |
| Maintenance (per drone)                  | £1,058             |
| Insurance / Communication (per drone)    | £1,000.00          |
| Total Battery Cost (per drone)           | £758.00            |
| Labour                                   | £10,000.00         |
| <b>Operating Costs (per drone)</b>       | <b>£12,868.79</b>  |

To ensure consistency across all design alternatives, R&D costs remain constant throughout the analysis. In the deterministic DCF evaluation for the medium drone fleet, the NPV is £1,349,620, with an IRR of 29.36%. This relatively strong IRR suggests that Alternative B is financially viable, offering favourable returns on investment.

When factoring in the time value of money with a discount rate of 12%, the payback period begins at approximately two years but extends significantly by year three, lagging one year behind Alternative A. While this difference is not substantial, it indicates that Alternative A may recover its initial investment more quickly, though Alternative B still remains an attractive option in terms of long-term profitability.



**Figure 6:** Cumulative Cash Flow for Alt B

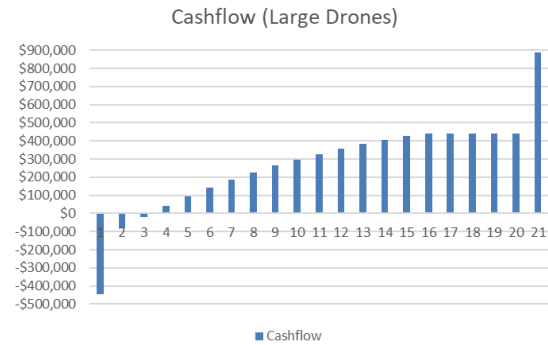
The medium drone fleet presents a strong long-term investment, offering higher payload capacity and extended range compared to the small drone fleet. While its financial performance remains competitive, the longer payback period and higher per-drone costs make it less attractive in the short term than Alternative A. However, for operations prioritising scalability and delivery flexibility, this option remains a viable contender, particularly in scenarios where larger payloads or longer delivery distances are required.

### 5.1.3 Large Drones - Aaron

The final deterministic scenario modelled the economic performance of the system, assuming a smaller network of large drones. Larger drones can stay flying for longer and carry heavier payloads, thanks to their increased battery capacity. This was translated into an increased delivery capacity per drone of 50 per day, leading to a lower size fleet of 15 drones.

**Table 5: Alt C Cost Parameters**

| Cost                                                         | Value              |
|--------------------------------------------------------------|--------------------|
| Individual Drone Manufacture & Installation Cost (per drone) | £13,000.00         |
| <b>Capacity cost (per drone at 115%)</b>                     | <b>£14,950.00</b>  |
| <b>Annual Infrastructure Costs</b>                           | <b>£446,760.00</b> |
| Yearly Battery Charge (per drone)                            | £15.64             |
| Maintenance (per drone)                                      | £1,808.00          |
| Insurance / Communication (per drone)                        | £2,300.00          |
| Total Battery Cost (per drone)                               | £1,508.00          |
| Labour (per drone)                                           | £10,000            |
| <b>Operating Costs (per drone)</b>                           | <b>£15,631.64</b>  |



**Figure 7: Cumulative Cash Flow for Alt C**

However, the increased performance of the drone leads to a significant CAPEX increase per drone. Specifically, the acquisition cost of one large drone is 41.18% more than a small one, and 20% more than a medium one. Although this cost is slightly recuperated from the increased revenue (13.64% more), it does not lead to an increase in overall economic viability. In fact, the large drones system has the worst performance, with an NPV of £626,028.47, and an IRR of 19.25%, more than 20% less than the small drones configuration. Also, the payback period is the highest out of the three options, taking 4 years to make meaningful returns.

In sum, under deterministic modelling, the large drones configuration is the worst investment alternative, primarily due to the CAPEX outweighing the increased revenue from a higher performing drone. Furthermore, as the fleet of large drones is smaller in size, it fails to capture the time-dependent demand of the food delivery market. Notably, in the simplified models analysed here, the daily peaks of demand is not taken into account. The system will experience the highest demand around lunch and dinner times [14], and therefore, there is inherent value in having a larger number of drones in the fleet to meet those intense periods of demand.

### 5.1.3 Comparison Evaluation

The deterministic DCF analysis determined that all alternatives were financially viable,

despite Alternative C having a significantly lower NPV and IRR.

- Alternative A had the highest NPV and IRR.
- Alternative B had slightly lower NPV and IRR values.
- Alternative C had the lowest NPV and IRR.

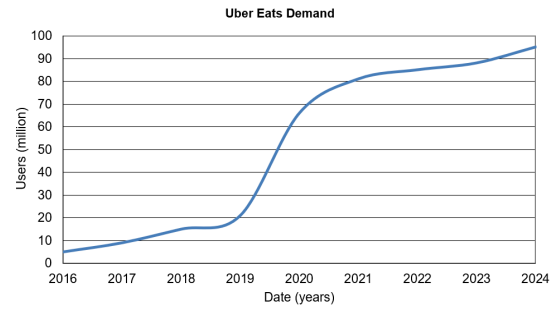
From a financial standpoint, Alternative A emerges as the most advantageous choice. Additionally, it offers the greatest flexibility for future expansion, as its individual drone manufacturing cost is the lowest. Furthermore, the drone itself is the smallest and requires the least space, making the deployment of additional drones and hub expansion more cost-effective.

## 6 Uncertainty Analysis

Engineering history shows that deploying systems with a purely deterministic approach can lead to catastrophic failures. The world is inherently uncertain, and rigid forecasts can distort economic models, misrepresenting viability. To enhance realism, it is crucial to incorporate reasonable uncertainty into the model. Three key sources of uncertainty were identified, selected based on their potential impact on overall economic performance: demand, cost driven by lithium-based batteries, and MARR sensitivity.

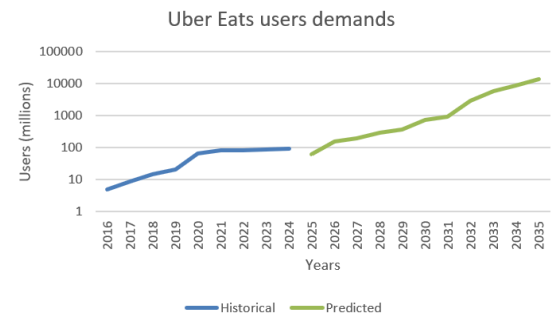
### 6.1 Demand Uncertainty

The demand for drone deliveries is one of the most crucial uncertain variables in our system. As our revenue model works on a per-delivery basis, the economic health of our system is closely related to the amount of trips we can complete. In the previous section, a constant growth rate was assumed. Using Uber Eats previous years demand, [15] a historical data analysis was carried out to find average growth rate and volatility, 49% and 68% respectively.



**Figure 8:** *Historical data*

This was then used to predict a possible forecast of increasing demand for delivery food services for the drone delivery system, which was what was used to simulate the growth rates used in the previous section.



**Figure 9:** *Combined demand*

However, in reality, uncertainties arise. The following demand uncertainty assumptions were used to generate our randomised NPV model to then be implemented in the Monte Carlo simulation:

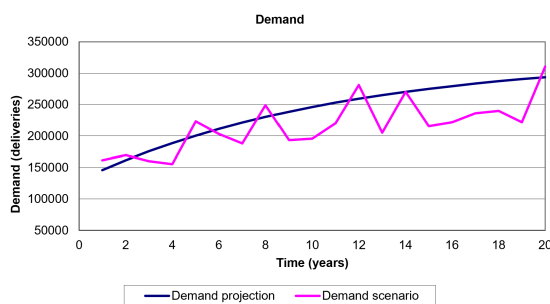
**Table 6:** *Uncertainty Assumptions*

|                                 | Level      | Why?                                                                                      |
|---------------------------------|------------|-------------------------------------------------------------------------------------------|
| Realised demand within Year 1   | 60% (high) | Initial adoption challenges, public perception challenges, and limited service areas [17] |
| Additional demand by Year 10    | 50% (med)  | Increase in widespread adoption, still some adoption challenges, competing tech           |
| Additional demand after Year 10 | 20% (low)  | Market Saturation, greater implementation into society                                    |
| Annual Volatility of demand     | 40%        | Using historical data of emerging tech industries, demand growth is highly                |

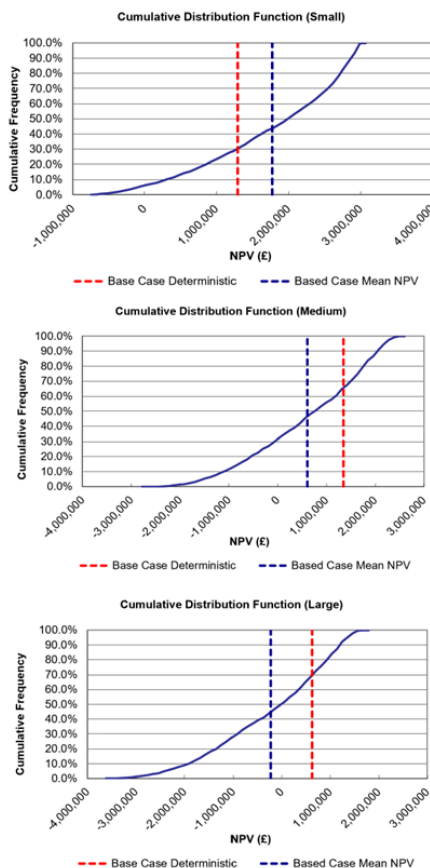
|        |  |                                                                             |
|--------|--|-----------------------------------------------------------------------------|
| growth |  | volatile due to regulatory, technological, and consumer adoption challenges |
|--------|--|-----------------------------------------------------------------------------|

Using these factors, the model was made more realistic, incorporating a wide range of NPV outcomes based on various possible demand scenarios influenced by the selected uncertainty values.

Below is a graph illustrating potential demand fluctuations compared to the initially projected demand, based on the Uber Eats historical analysis model.



**Figure 10: Demand forecasts**



**Figure 11: Uncertainty model comparisons**

By implementing these uncertainties into all three alternative models, the medium and large drone alternatives had a Mean NPV lower than the Base Case Deterministic NPV, indicating they are riskier alternatives.

Consequently, the analysis concluded that, under these uncertainties, small drones emerged as the most robust choice.

## 6.2 Lithium Price

Since the small drones were identified as the most robust option in the demand uncertainty analysis, we proceeded with this configuration for the lithium price uncertainty analysis.

Over the past three years, lithium prices have experienced extreme volatility, peaking in 2022 before declining sharply [18]. This fluctuation is particularly significant for the drone delivery industry, as lithium-based batteries are a major cost driver in fleet operations. Raw materials and components can constitute up to 50-60% of total manufacturing costs. Within this category, batteries account for approximately 20% of the raw material expenses. This implies that battery costs can represent about 10-12% of the total manufacturing costs of a drone. [19]

Analysing historical data (Figure 12) reveals an average growth rate of 0.01% and a volatility rate of 13.78%, highlighting the uncertainty in long-term battery costs. Given that battery expenses directly impact operating costs and profitability, understanding future lithium price trends is crucial for assessing the financial stability and scalability of the system.

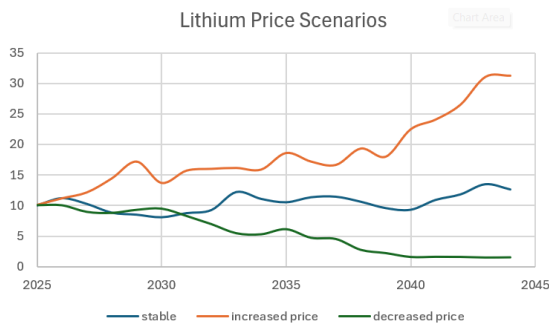


**Figure 12:** Average Price of Lithium per kg

Three lithium price scenarios were simulated to assess their impact on the small drone fleet's NPV:

1. **Stable Lithium Prices:** Lithium prices remain constant with minimal fluctuation, resulting in steady battery costs and predictable operating expenses.
2. **Volatile Price Increase:** A sharp rise in lithium prices increases battery costs, leading to higher operational expenses and potentially lower profitability.
3. **Volatile Price Decrease:** A decline in lithium prices reduces battery costs, improving operating margins and increasing NPV.

These scenarios, shown in Figure 13, allow us to evaluate the sensitivity of the NPV model to price fluctuations and determine the financial resilience of the small drone fleet under different market conditions.



**Figure 13:** Lithium Price Scenarios

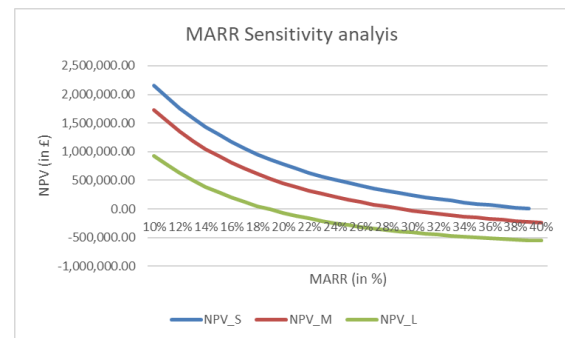
After modelling the impact of lithium price fluctuations under different scenarios and inputting the price projections into the average

operating costs for the small drone fleet, the fluctuations observed, had minimal impact on the overall financial performance. Despite some variations in battery costs, the NPV and overall profitability of the small drone fleet remained relatively stable across all scenarios.

Based on these results, we concluded that price fluctuations in lithium are unlikely to significantly affect the long-term economic viability of the system. While monitoring lithium price trends is still important, the system appears to be resilient to price changes, making it a robust investment option under various market conditions.

### 6.3 Sensitivity Analysis

Other sources of uncertainty, such as weather conditions and the implementation of restrictive legislations, were included in the conservative MARR. The sensitivity analysis of MARR on NPV for three alternatives is shown in Figure 14, highlighting how each alternative responds to changes in economic and external factors.



**Figure 14:** MARR Sensitivity Analysis Graph

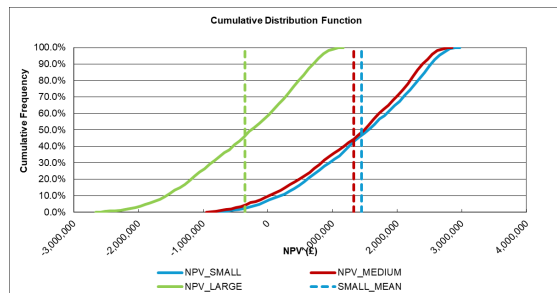
1. Alternative A (blue) is the most sensitive to MARR fluctuations, with a significant drop in NPV after 36%, indicating it is the least resilient.
2. Alternative B (red) has a moderate sensitivity, with the NPV remaining positive until around 38%, showing more stability than NPV\_S.

3. Alternative C (green) is the most resilient, maintaining a positive NPV up to a MARR of 40%, making it the least affected by MARR changes.

Overall, Alternative A is the most vulnerable, while Alternative C is the most advantageous due to the increased number of deliveries.

## 6.4 Monte Carlo Simulation

By incorporating Demand Uncertainty as well as the minimal effects of Lithium market uncertainties, a Monte Carlo Simulation was carried out. A sample size of 2000 was used to capture the stochastic behaviour of the three systems.



**Figure 15:** Monte Carlo Simulation Graph

To further compare the randomised NPV models of the three drone sizes, their key metric summary accounting for uncertainties was compared.

**Table 7:** Summary Table with Uncertainty

| Metric              | Small         | Medium         | Large          |
|---------------------|---------------|----------------|----------------|
| Mean NPV            | £1,773,951.71 | £607,574.44    | -£225,728.87   |
| Value at risk (P5)  | -£96,291.56   | -£1,511,431.77 | -£2,365,940.65 |
| Value at Gain (P95) | £2,936,298.76 | £2,185,447.14  | £1,353,523.64  |
| Stand Dev.          | £971,208.52   | £1,182,171.65  | £1,189,355.12  |
| Initial CAPEX       | £521,400.00   | £585,108.00    | £671,010.00    |

## 6.5 Evaluation

The small drone alternative proves to be the most robust choice when accounting for demand uncertainty and lithium price volatility. Monte Carlo simulations show that small drones maintain a stable NPV, even with fluctuating demand and lithium price changes. In contrast, medium and large drones exhibited higher risk, with lower NPVs in randomised simulations.

Lithium price fluctuations, while impactful on operating costs, had minimal effect on the small drone fleet's overall profitability. Additionally, the small drone model showed more resilience to changes in MARR compared to the other alternatives, maintaining a positive NPV even under fluctuating economic conditions.

Ultimately, considering all the uncertainties evaluated—demand, lithium price fluctuations, and MARR sensitivity—the small drone alternative presents the most balanced risk-to-reward profile, making it the optimal choice for the drone delivery system.

## 7 Flexibility Analysis

### 7.1 Decision rules

To assess the value of system flexibility, we have outlined three distinct decision rules designed to leverage the best design alternative at this stage: the small drones network. These rules account for the gap between capacity and demand from the previous year, guiding the decision on whether to deploy an additional drone into the network.

- **Risky Decision rule:** If the difference between demand and capacity is greater than  $x0.25$  the delivery capacity of one drone, **then** deploy an additional drone into the network, **else** keep the number of drones constant.

- **Pragmatic** Decision rule: **If** the difference between demand and capacity is *greater* than the delivery capacity of one drone, **then** deploy an additional drone into the network, **else** keep the number of drones constant.
- **Conservative** Decision rule: **If** the difference between demand and capacity is greater than  $x1.5$  the delivery capacity of one drone, **then** deploy an additional drone into the network, **else** keep the number of drones constant.

## 7.2 Evaluation

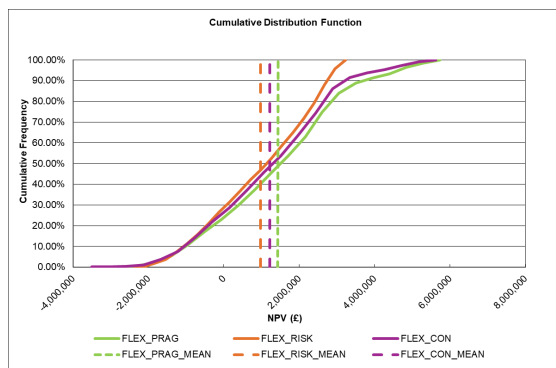


Figure 16: CDF of different flexible scenarios

Table 8: Recap of different flexible scenarios

| Metric              | Risky      | Pragmatic  | Conservative |
|---------------------|------------|------------|--------------|
| Mean NPV            | £1,358,54  | £1,779,96  | £1,578,62    |
| Value at risk (P5)  | £-146,425  | £-100,139  | £96,291.56   |
| Value at Gain (P95) | £3,200,331 | £3,010,386 | £2,801,571   |
| Stand Dev.          | £1,046,782 | £953,031   | £885,721     |

## 6.5 Evaluation

In Figure 16 above, the CDF for all three flexible scenarios confirms that the optimal flexibility strategy is the “pragmatic” one, as it leads to the highest mean NPV compared to the “risky” and “conservative” models. The

“pragmatic” decision rule surpasses the others for two main reasons:

1. The system adjusts drone deployment based on demand. If demand decreases, no additional drones are deployed. If demand increases, extra drones are added to maximise efficiency and ensure timely deliveries.
2. This approach balances risk and reward. A high risk strategy may lead to volatility, while a “conservative” strategy might miss revenue opportunities. By responding to demand signals and deploying drones strategically, the system reduces risk while optimising returns.

## 8 Discussion

Recommendation on the best Design:

- The small drone network is the optimal design for the system, offering the most robust performance under demand, and lithium price uncertainties.
- The pragmatic decision rule is identified as the best flexibility strategy, as it adjusts drone deployment based on demand fluctuations, balancing risk and reward. This approach results in the highest mean NPV by maximising efficiency and profitability.

Limitations of the work include:

- The analysis assumes consistent demand growth and does not fully account for potential disruptions such as regulatory changes or market competition.
- Sensitivity to extreme economic shifts or unexpected external factors is not fully captured.

- The pragmatic decision rule is based on historical data, and future demand trends may differ from the assumptions, highlighting the need for ongoing adjustments and real-world validation.

## 9 Conclusion

After a detailed DCF evaluation of three investment options for a drone delivery network, factoring in uncertainty and system flexibility, we found that in every case a

smaller, more adaptable system is preferable. This minimises upfront investment while maximising returns. This finding aligns with the strategies of modern drone logistic firms, that prioritise small hubs spread out in different cities. Furthermore, this approach allows companies to continuously update their core technology offering, deploying their designs as time goes on.

Companies embracing flexible drone fleets to respond to demand shifts operate under this philosophy, and our analysis confirms its financial viability in an unpredictable market.

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# 11 Appendix

Link to excel sheets: [Economics & Finance Group - ND | General | Microsoft Teams](#)